

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE CODE: ITA 06

COURSE NAME: MACHINE LEARNING

COURSE OUTCOME COVERED

CO3: Bayes Theorem – Concept Learning – Maximum Likelihood – Minimum Description Length Principle – Bayes Optimal Classifier – Gibbs Algorithm – Naïve Bayes Classifier – Bayesian Belief Network – EM Algorithm – Probability Learning – Sample Complexity – Finite and Infinite Hypothesis Spaces – Mistake Bound Model, (BL 6)

Assessment Tool Weightage: 20%

QUESTIONS: 15

Total Marks: 25

Annexure A

Case Study

Code of Conduct:

I V.Madhulika (192312620) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

V.Madhulika

Signature of the student

1. A hospital aims to develop an intelligent system to diagnose diseases based on patient symptoms and medical history. Using concepts such as Bayes' Theorem, Naïve Bayes classifiers, and Bayesian Belief Networks, design a probabilistic model for diagnosis. Analyze how uncertainty is handled and how conditional dependencies between symptoms influence predictions. Discuss the role of probability learning and evaluate the effectiveness of your approach with respect to real-world constraints.

Answer:

Model Design:

Symptoms and medical history are represented as features. A Naïve Bayes classifier serves as a fast baseline, assuming symptoms are conditionally independent given the disease and computing $P(\text{Disease} \mid \text{Symptoms}) \propto P(\text{Symptoms} \mid \text{Disease}) \cdot P(\text{Disease})$. Where symptoms have known dependency structure (e.g., fever co-occurring with infection markers), a Bayesian Belief Network is used instead, explicitly encoding conditional dependencies between symptoms and diseases through a directed graph with conditional probability tables (CPTs).

Handling Uncertainty & Dependencies:

Bayes' Theorem represents diagnostic uncertainty as a full probability distribution over possible diseases rather than a single hard label, updating prior disease prevalence with observed symptom evidence to produce posterior probabilities. The BBN additionally captures how symptoms reinforce each other — e.g., cough and fever together increasing the likelihood of a respiratory infection more than either alone — something the independence-assuming Naïve Bayes model cannot represent.

Role of Probability Learning:

Prior probabilities (disease prevalence) and conditional probability tables are estimated from historical patient records and continually refined as new patient data becomes available.

Evaluation:

Naïve Bayes is simple and fast but can misestimate probabilities when symptoms are strongly correlated; BBNs represent true dependencies more accurately but require more data and computation to learn reliable CPTs. This hybrid probabilistic approach supports earlier diagnosis while explicitly communicating uncertainty (e.g., '70% likelihood of Condition A') to clinicians. Real-world constraints — incomplete patient records and rare diseases with sparse data — limit reliability and require careful smoothing and expert-informed priors.

2. An organization wants to build a spam detection system for emails using machine learning techniques. Using Naïve Bayes and Bayes Optimal Classifier concepts, design a classification system. Compare Maximum Likelihood estimation with Minimum Description Length (MDL) principle in feature selection. Analyze system performance, discuss trade-offs in model complexity, and evaluate how sample size and hypothesis space affect classification accuracy.

Answer:

Model Design:

Each email is represented as a Bag-of-Words/TF-IDF feature vector. A Multinomial Naïve Bayes classifier estimates $P(\text{word} \mid \text{Spam})$ and $P(\text{word} \mid \text{Not Spam})$ from labeled training emails and classifies new emails via Bayes' Theorem. The Bayes Optimal Classifier concept is used as the theoretical benchmark — averaging predictions over the hypothesis space weighted by posterior probability — against which the practical Naïve Bayes approximation is compared.

MLE vs MDL in Feature Selection:

MLE selects probability estimates that maximize the likelihood of the observed training emails, which can lead to overfitting on rare, spurious words. The MDL principle instead favors the simplest feature set/model that still explains the data well, trading a small amount of fit for a more compact, generalizable model — useful for pruning noisy, uninformative words from the feature set.

Performance & Trade-offs:

A model selected via MDL typically generalizes better with limited training data but may miss minor signal words; a model fit purely by MLE with many features can achieve high training accuracy but overfit, particularly with a small training set relative to feature-set size.

Effect of Sample Size & Hypothesis Space:

With few training emails, only a small, simple hypothesis (feature set) can be reliably estimated; as data grows, richer feature sets (a larger hypothesis space) become learnable without overfitting, generally increasing accuracy.

3. A retail company wants to segment its customers based on purchasing behavior, but the data contains hidden patterns and incomplete information. Apply the EM algorithm to identify clusters and estimate underlying distributions. Analyze convergence behavior, initialization challenges, and the role of probability learning. Discuss how the results can be used for targeted marketing and evaluate limitations of the approach.

Answer:

Model Design:

Customer purchasing behavior is modeled as a Gaussian Mixture Model where each mixture component represents a hidden customer segment. EM iteratively estimates segment-membership probabilities (E-step) and updates each segment's distribution parameters (M-step) despite incomplete purchase records.

Convergence & Initialization:

Convergence is monitored via the change in log-likelihood between iterations. Poor random initialization of segment centers can cause slow convergence or convergence to a local optimum representing a suboptimal segmentation — mitigated by multiple random restarts or K-Means-based initialization.

Role of Probability Learning:

EM learns the underlying probability distribution of each segment directly from incomplete data, treating true segment membership as a latent (hidden) variable estimated jointly with the distribution parameters.

Application & Limitations:

The resulting soft cluster assignments (probability of belonging to each segment) enable targeted marketing — e.g., customers with a high probability of belonging to a 'high-value' segment receive premium offers. Limitations include sensitivity to the assumed number of clusters, the Gaussian distributional assumption not always matching real purchasing patterns, and reduced reliability when a large fraction of the data is missing.

4. A research team is developing a concept learning system to classify medical images into normal and abnormal categories. Using hypothesis space search and the Candidate Elimination approach, design a model that learns from labeled examples. Analyze the impact of finite vs infinite hypothesis spaces, sample complexity, and mistake bound model on learning performance. Evaluate how well the system generalizes to unseen data.

Answer:

Model Design:

Each labeled image example is represented as a feature vector (e.g., presence/intensity of specific patterns). Candidate-Elimination maintains the version space — the Specific (S) and General (G) boundary sets of hypotheses consistent with all labeled examples seen so far — converging toward a hypothesis that distinguishes normal from abnormal images.

Hypothesis Space Impact:

A finite hypothesis space (e.g., discretized feature values) guarantees the version space will eventually converge given enough examples, with sample complexity bounded by the size of the space; an infinite hypothesis space (e.g., continuous-valued image features) requires additional structure — such as bounded complexity/VC-dimension — to guarantee convergence and reliable generalization.

Mistake Bound Model:

Provides a worst-case bound on the number of misclassifications the learner can make before converging to a correct hypothesis, useful for estimating how many labeled images are needed before the system becomes reliable in an online learning setting.

Generalization Evaluation:

The learned hypothesis is evaluated on a held-out set of unseen images. Performance depends heavily on how well the chosen hypothesis representation (inductive bias) matches the true decision boundary between normal and abnormal cases — an overly restrictive hypothesis space may fail to represent the true concept (underfitting), while an overly large one requires substantially more data to avoid ambiguity.

5. A financial institution needs to make investment decisions under uncertain market conditions. Using Bayesian inference, Gibbs algorithm, and probability learning, design a system to predict outcomes and update beliefs dynamically. Analyze how prior knowledge influences predictions, evaluate risk using posterior probabilities, and discuss how the model adapts as new data becomes available. Consider computational challenges and optimization strategies.

Answer:

Model Design:

Bayesian inference maintains a probability distribution over market states/outcomes, updating beliefs as new market data arrives via Bayes' Theorem. The Gibbs algorithm samples a hypothesis (e.g., a specific market model) from the posterior distribution over models rather than committing to one point estimate, better reflecting genuine model uncertainty. Probability learning continually estimates outcome probabilities (e.g., asset price movement) from historical and streaming data.

Prior Influence:

Prior beliefs (historical average returns, expert judgment) meaningfully shape early predictions when data is limited, gradually giving way to data-driven posterior estimates as more market observations accumulate.

Risk Evaluation:

Posterior probability distributions over outcomes directly support risk assessment — e.g., computing the probability that a return falls below a threshold — enabling more informed, uncertainty-aware investment decisions than single-point forecasts.

Adaptation & Computational Challenges:

The model updates dynamically as new data streams in (online Bayesian updating). Computational challenges include maintaining and sampling from complex posteriors in real time, addressed through approximate inference techniques (e.g., Markov Chain Monte Carlo methods, of which Gibbs sampling is one) and periodic re-estimation rather than full recomputation.

6. An autonomous vehicle system must make real-time driving decisions under uncertain environmental conditions such as weather, traffic, and pedestrian movement. Using Bayesian inference, Naïve Bayes classifiers, and probabilistic reasoning, design a model to predict safe driving actions. Analyze how uncertainty and sensor noise are handled, and evaluate system reliability under dynamic conditions.

Answer:

Model Design:

Bayesian inference fuses and updates beliefs from multiple noisy sensors (camera, radar, lidar), while Naïve Bayes classifiers provide quick per-sensor event classification (e.g., 'pedestrian present' given detected features). These feed into a probabilistic reasoning module that outputs the safest driving action given the current belief state.

Uncertainty & Noise Handling:

Sensor noise is modeled explicitly as a likelihood/measurement-error distribution (e.g., probability of a true pedestrian given a noisy detection signal). Bayesian updating combines multiple noisy, partially independent

sensor readings to produce a more reliable posterior belief than any single sensor alone, reducing the impact of individual sensor errors.

Reliability Evaluation:

System reliability is evaluated under simulated adverse conditions (fog, rain, sensor dropout) by measuring how quickly and accurately posterior beliefs about hazards converge to ground truth. A well-calibrated probabilistic model degrades gracefully — producing appropriately higher uncertainty rather than confidently making wrong decisions — when sensor data is poor, which is critical for safety-critical real-time deployment.

7. A healthcare organization aims to predict patient risk levels for chronic diseases using historical and lifestyle data. Apply probability learning, Bayes' Theorem, and hypothesis space search to design the model. Analyze the impact of prior probabilities, feature dependencies, and sample size. Evaluate the model's interpretability and its usefulness in preventive healthcare strategies.

Answer:

Model Design:

Bayes' Theorem combines prior disease prevalence with patient-specific likelihoods derived from historical/lifestyle data (diet, exercise, family history). Hypothesis space search selects an appropriate risk-scoring model consistent with observed outcomes across the patient population, while probability learning estimates the conditional probabilities of risk factors from cohort data.

Impact of Priors, Dependencies & Sample Size:

Prior probabilities (baseline population disease rate) strongly influence predictions when individual patient data is limited. Correlated features (e.g., BMI and blood pressure) can bias probability estimates if treated as independent, so dependency structure must be considered. Larger sample sizes yield more statistically reliable conditional probability estimates, particularly for rarer risk-factor combinations.

Interpretability & Usefulness:

A probabilistic risk score (e.g., '35% five-year risk of Type 2 Diabetes') is interpretable to both clinicians and patients, supporting preventive interventions such as lifestyle counseling and earlier screening targeted at high-risk individuals — though priors must reflect the target population to avoid systematic bias.

8. A bank wants to detect fraudulent transactions using machine learning under uncertain and evolving patterns. Using Bayesian Belief Networks and EM algorithm concepts, design a detection system. Analyze hidden variables, incomplete data handling, and convergence issues. Evaluate how probability-based reasoning improves detection accuracy and reduces false positives in real-world scenarios.

Answer:

Model Design:

Transaction features, hidden fraud-relevant factors, and the outcome (fraud/legitimate) are represented as nodes in a Bayesian Belief Network with learned CPTs. The EM algorithm estimates parameters when some variables (e.g., true fraud label in older data, or latent behavioral patterns) are missing or only partially observed.

Hidden Variables & Incomplete Data:

EM treats unobserved fraud indicators as latent variables, iteratively estimating their expected values (E-step) and re-estimating model parameters (M-step) to best explain observed transaction patterns despite incomplete labeling.

Convergence Issues:

Poor initialization or overlapping fraud/legitimate distributions can cause slow convergence or convergence to a local optimum that under- or over-estimates fraud likelihood; this is monitored via log-likelihood stabilization and mitigated with multiple restarts.

Evaluation:

Probability-based reasoning outputs a fraud probability rather than a hard decision, letting analysts set a threshold balancing detection recall against the false positives that inconvenience legitimate customers. This calibrated uncertainty is generally more actionable in dynamic fraud environments than a black-box binary classifier.

9. An e-commerce platform aims to recommend products to users based on browsing and purchase history. Using Naïve Bayes and Bayesian learning concepts, design a recommendation model. Analyze how conditional probabilities influence recommendations and how uncertainty in user preferences is managed. Evaluate scalability and performance in handling large, dynamic datasets.

Answer:

Model Design:

Naïve Bayes estimates the probability a user will engage with a given product category given features from their browsing/purchase history; broader Bayesian learning combines multiple weak behavioral signals (recency, purchase frequency, category preference) into a posterior preference estimate used for ranking recommendations.

Conditional Probabilities & Uncertainty:

Conditional probabilities such as $P(\text{purchase category } X \mid \text{previously viewed category } Y)$ directly drive which items are recommended. Uncertainty in user preference — particularly for a new user with sparse history (the 'cold start' problem) — is handled by relying more heavily on prior/population-level probabilities initially, gradually shifting weight toward individual behavior as more interactions are observed.

Scalability & Performance:

For large, dynamic catalogs, probability tables must be updated incrementally (online learning) rather than recomputed from scratch. Naïve Bayes' computational simplicity — independent per-feature probability estimates — makes it well suited to scale to large numbers of users/items compared to more complex joint-probability models, though at some cost to capturing complex cross-item dependencies.

10. A research institute is building a system to predict climate changes using historical environmental data. Apply probabilistic models such as Bayesian inference and EM algorithm to handle incomplete and noisy datasets. Analyze convergence behavior, uncertainty modeling, and the influence of prior knowledge. Evaluate the system's effectiveness in long-term environmental forecasting.

Answer:

Model Design:

Bayesian inference combines historical climate records (prior knowledge) with new environmental observations to update predictions of future climate variables. The EM algorithm handles incomplete/noisy historical records (e.g., missing sensor readings from certain decades or regions) by estimating latent 'true' environmental states.

Convergence & Uncertainty Modeling:

EM convergence is tracked via stabilization of estimated climate parameters across iterations. Bayesian credible intervals around predictions explicitly represent uncertainty due to noisy/incomplete historical data, which is essential given the long time horizons and limited historical measurement coverage typical of climate data.

Prior Knowledge Influence:

Well-established physical/climatological priors (e.g., known greenhouse-gas relationships) stabilize estimates especially where direct historical data is sparse or noisy, though incorrect priors could bias long-term forecasts if not periodically validated against new data.

Effectiveness:

Such probabilistic models produce forecasts with quantified confidence rather than single deterministic numbers, aiding long-term planning — though genuine long-term forecasting accuracy remains limited by the inherent complexity and non-stationarity of climate systems.

11. A telecommunications company aims to predict customer churn using historical usage data and service feedback. Using probabilistic models such as Naïve Bayes, Bayesian inference, and hypothesis space search, design a predictive system. Analyze how prior probabilities and feature dependencies influence churn prediction. Evaluate how uncertainty in customer behavior is handled and discuss the effectiveness of the model in improving customer retention strategies.

Answer:

Model Design:

Naïve Bayes and broader Bayesian inference estimate $P(\text{Churn} \mid \text{usage features, feedback})$ from historical customer records; hypothesis space search identifies which combination of features (usage decline, complaint frequency, contract type) best predicts churn.

Prior & Dependency Influence:

The prior churn rate (base rate) anchors predictions, especially for customers with limited history. Feature dependencies — e.g., low usage combined with a recent complaint jointly indicating much higher churn risk than either alone — are only approximately captured under the Naïve Bayes independence assumption, so a more dependency-aware model may improve accuracy at the cost of added complexity.

Uncertainty Handling:

The model outputs a churn probability per customer rather than a hard label, letting the business rank customers by risk and prioritize retention efforts (special offers, proactive support) toward the highest-probability cases.

Effectiveness:

Probabilistic churn scores integrate well with targeted retention campaigns, improving the efficiency of retention spend, though model accuracy depends on how representative and complete the historical usage/feedback data is, and the model should be periodically retrained as customer behavior evolves.

12. A smart healthcare system is designed to monitor patients in real-time using wearable sensors. Using Bayesian Belief Networks and probability learning, develop a model to detect anomalies in vital signs. Analyze how conditional dependencies between physiological parameters are modeled and how uncertainty due to sensor noise is handled. Evaluate system performance and discuss challenges in real-time deployment.

Answer:

Model Design:

A Bayesian Belief Network links physiological parameters (heart rate, blood pressure, oxygen saturation) as interdependent nodes, with an 'anomaly' node whose posterior probability updates in real time as new sensor readings (evidence) arrive; probability learning continuously refines the CPTs from accumulated patient data.

Dependency & Noise Modeling:

Conditional dependencies capture realistic physiological relationships — e.g., elevated heart rate combined with low oxygen saturation being jointly more indicative of distress than either alone. Sensor noise is represented as a likelihood/measurement-error distribution, so an isolated noisy reading is weighed against the broader pattern of evidence rather than triggering a false alarm on its own.

Evaluation & Deployment Challenges:

Performance is evaluated by detection sensitivity (catching true anomalies) versus false-alarm rate under simulated noisy sensor conditions. Real-time deployment challenges include the computational cost of continuous inference, gracefully handling sensor dropout/missing readings, and adapting the network to individual patient baselines rather than relying on a fixed population-level model.

13. A logistics company wants to optimize delivery routes under uncertain traffic and weather conditions. Using Bayesian inference and probabilistic reasoning, design a system that predicts optimal routes dynamically. Analyze how prior knowledge and real-time data updates influence decision-making. Evaluate computational complexity and discuss how uncertainty impacts route optimization efficiency.

Answer:

Model Design:

Bayesian inference maintains and updates a probability distribution over expected traffic/weather-related delays for candidate routes, incorporating both historical patterns (prior) and real-time updates (current traffic, weather alerts) to dynamically re-rank routes by expected delivery time.

Prior & Real-Time Updating:

Historical average conditions (prior) provide an initial route ranking before a trip begins; as real-time data arrives, posterior estimates are updated, allowing dynamic rerouting when conditions change significantly from expectations.

Computational Complexity & Efficiency:

Maintaining and updating posterior distributions across many possible routes in real time is computationally demanding, so practical systems typically restrict inference to a limited candidate set of routes and use efficient approximate updating rather than full recomputation.

Impact of Uncertainty:

By explicitly modeling delay uncertainty rather than a single expected value, the system can favor routes with a lower risk of significant delay — not merely the lowest expected delay — improving the reliability of delivery time estimates for customers.

14. A cybersecurity system is developed to detect network intrusions using probabilistic learning methods. Apply Naïve Bayes classifiers, EM algorithm, and Bayesian learning to model normal and abnormal network behavior. Analyze how hidden patterns and incomplete data are handled. Evaluate the system's effectiveness in reducing false alarms and improving detection accuracy in dynamic environments.

Answer:

Model Design:

Naïve Bayes classifiers model the probability of network traffic being normal versus abnormal based on observed traffic features; the EM algorithm learns from partially labeled data (much network traffic is unlabeled) by treating normal/abnormal status as a latent variable where labels are missing; broader Bayesian learning continually updates the model as attack patterns evolve.

Hidden Patterns & Incomplete Data:

EM allows the model to discover latent structure — e.g., previously unseen attack sub-types — in largely unlabeled traffic logs, iteratively refining the estimated probability that each traffic pattern belongs to a normal versus intrusion category.

Effectiveness Evaluation:

A probabilistic approach outputs a graded intrusion-likelihood score rather than a hard alert, letting security analysts tune the decision threshold to balance detection recall against the excessive false alarms that burden security operations centers. Performance is evaluated via detection rate and false-alarm rate under simulated/evolving attack scenarios, with the model retrained periodically to adapt to newly observed attack patterns.

15. An agriculture-based system aims to predict crop yield under uncertain environmental conditions such as rainfall, soil quality, and temperature. Using Bayesian inference and probability learning, design a predictive model. Analyze how prior agricultural knowledge and seasonal variations influence predictions. Evaluate the model's reliability and discuss its impact on decision-making for farmers and policy planning.

Answer:

Model Design:

Bayesian inference combines prior agricultural knowledge (typical yield ranges for given soil/climate conditions) with observed seasonal data (rainfall, soil quality, temperature) to produce a posterior distribution over expected crop yield; probability learning continually updates yield-influencing probabilities as new seasonal outcomes are recorded.

Prior Knowledge & Seasonal Variation:

Prior agronomic knowledge (e.g., known optimal rainfall ranges for a crop) anchors predictions especially early in a growing season when current-season data is limited; as the season progresses and more real-time environmental data becomes available, posterior yield estimates are refined and become more locally accurate.

Reliability Evaluation:

The model's reliability is evaluated by comparing predicted yield distributions against actual harvest outcomes across multiple seasons, with well-calibrated uncertainty bounds — not just a single number — being especially valuable for risk-averse planning.

Decision-Making Impact:

Probabilistic yield forecasts, including uncertainty ranges, help farmers plan resource allocation (irrigation, fertilizer) and help policymakers plan food security and pricing strategies more robustly than single-point estimates, particularly under increasingly variable climate conditions.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Conceptual Understanding	30	Demonstrates clear and complete understanding of the concept	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor or incorrect understanding
Accuracy of Answer	20	Completely correct answer with no errors	Mostly correct with minor errors	Partially correct with noticeable errors	Incorrect or irrelevant answer
Application of Knowledge	20	Correctly applies concepts to solve the question/problem	Applies concepts with minor mistakes	Limited or partially correct application	No or incorrect application
Completeness of Response	15	Covers all required parts of the question	Covers most parts with minor omissions	Covers only some parts	Incomplete or missing response
Clarity & Presentation	15	Clear, concise, and well-organized answer	Understandable with minor issues	Somewhat unclear or poorly structured	Difficult to understand